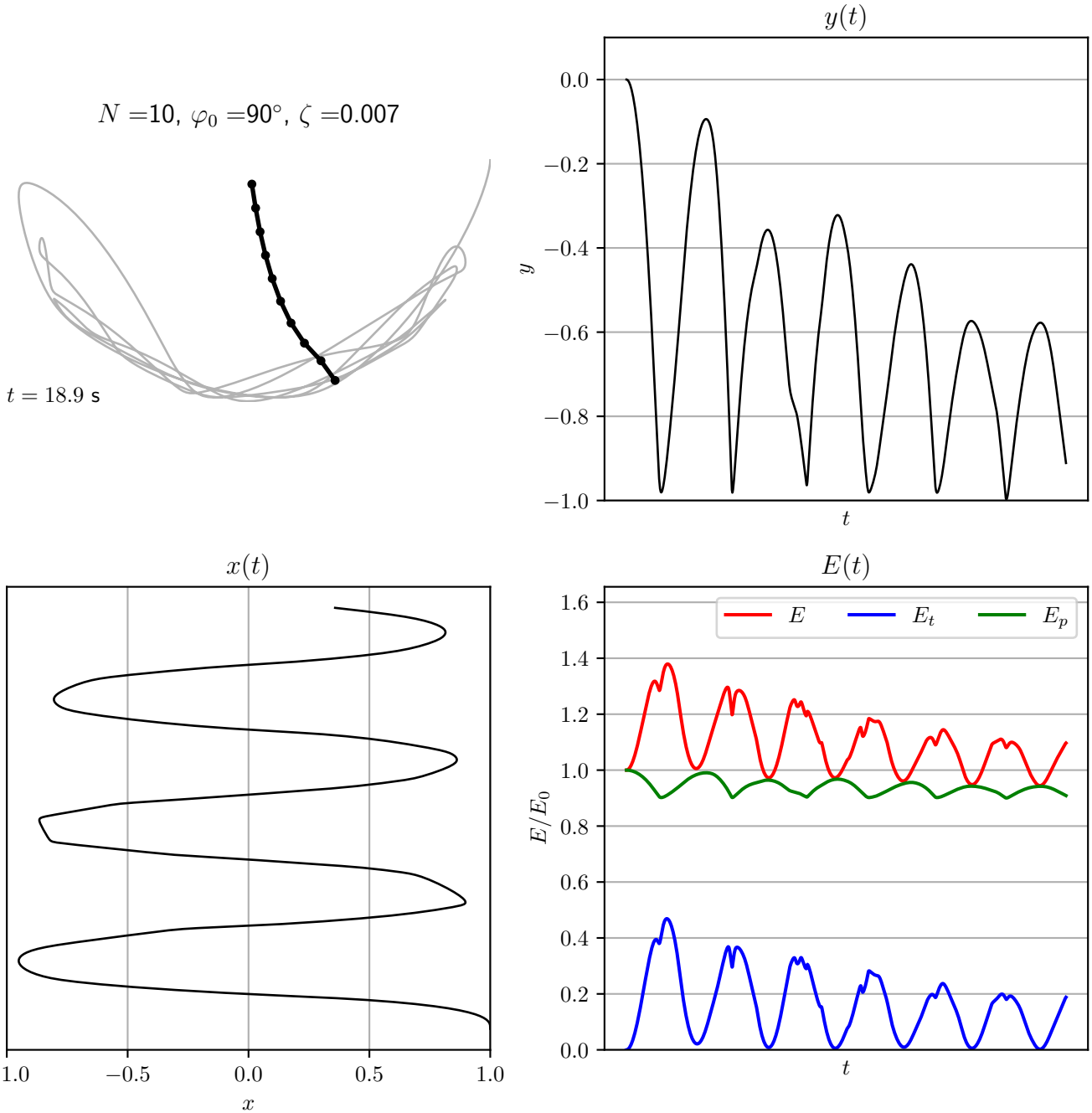

$N$ =10, $\varphi_0$ =90°, $\zeta$ =0.007
$t$ = 18.9 s
$y(t)$
$y$
$t$
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
$x(t)$
$t$
$x$
−1.0
−0.5
0.0
0.5
1.0
$E(t)$
$E$
$E_t$
$E_p$
$E/E_0$
$t$
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0